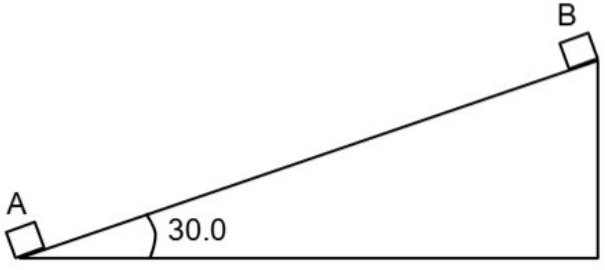
5 Two small masses A and B are placed at the top and bottom of a smooth slope inclined at 30.0° to the horizontal. At time $t = 0$, mass A is launched up the slope with an initial speed of 4.00 m s ^{1} while mass B is released from rest at the top of the slope. The two masses collide at a point along the slope.



The length of the slope is 5.00 m.

At what time does the collision occur?

A 0.681 s **B** 1.25 s **C** 1.50 s **D** 2.46 s